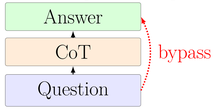
Markovian Transformers for Informative Language Modeling

Scott W. Viteri Max Lamparth Peter Chatain Clark Barrett
Department of Computer Science, Stanford University

ICLR 2026

Problem: CoT Is Not Load-Bearing

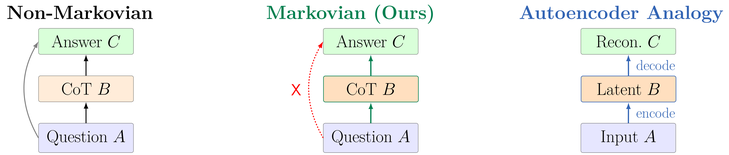
Standard Chain-of-Thought can be altered without changing the answer, because the model still sees the question when generating one.



Goal: not full faithfulness, but **informativeness**—the CoT should suffice to support the answer on its own.

Markovian Language Model

$M = (\mathcal{O}, \mathcal{S}, \pi, u, s_1)$: π predicts observations **from state alone**.



All information from A to C must pass through bounded-length B: an **autoencoder-style reasoning bottleneck**.

Informativeness Objective & Loss

Reward (trained vs. frozen baseline):

$$R_\theta = \ln \pi_\theta(\text{ans} \mid \text{CoT}) - \ln \pi'(\text{ans} \mid \text{CoT}') \quad ; \quad J(\theta) = \mathbb{E}_\tau[R_\theta(\tau)]$$

Gradient (two terms because R_θ depends on θ):

$$\nabla_\theta J = \mathbb{E}\Big[\underbrace{R_\theta \nabla_\theta \ln P_\theta}_{\text{policy gradient}} + \underbrace{\nabla_\theta R_\theta}_{\text{actor-reward}}\Big]$$

Loss: $\mathcal{L} = \mathcal{L}_{\text{PG}} + \mathcal{L}_{\text{AR}} + \mathcal{L}_{\text{KL}}$

$$\mathcal{L}_{\text{PG}} = -\ln u_\theta(\text{CoT} \mid q) \cdot A^{\text{detach}} \qquad \text{(policy gradient)}$$
$$\mathcal{L}_{\text{AR}} = -A \qquad \text{(actor-reward)}$$
$$\mathcal{L}_{\text{KL}} = \beta_{\text{KL}} D_{\text{KL}}(u_\theta \,\|\, u') \qquad \text{(regularization)}$$

Training Algorithm

Given: dataset P of (q, a), actor (u_θ, π_θ), frozen baseline (u', π')
for each training batch of size B **do**
  Sample $(q, a) \sim P$
  Sample $\text{CoT}_i \sim u_\theta(\cdot \mid q, \text{CoT}_{\text{init}})$ for $i = 1..B$
  Sample $\text{CoT}' \sim u'(\cdot \mid q, \text{CoT}_{\text{init}})$ (frozen baseline)
  $r_i = \ln \pi_\theta(a \mid \text{CoT}_i)$; $b = \ln \pi'(a \mid \text{CoT}')$
  $R_i = r_i - b$; standardize: $A_i = (R_i - \mu)/(\sigma + \epsilon)$
  Compute $\mathcal{L}_{\text{PG}}$, $\mathcal{L}_{\text{AR}}$, $\mathcal{L}_{\text{KL}}$; update θ
end for

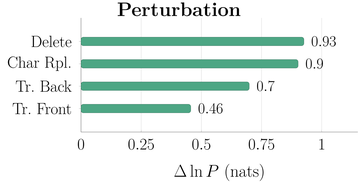
Main Result: Accuracy vs. Fragility

Markovian training trades a few accuracy points for CoTs the model actually relies on.

Accuracy

Task	Base	NM	Mkv
GSM8K	19.6	63.3	57.1
ARC	36.1	78.6	79.9
Arith.	1.0	97.0	98.0
MMLU	21.4	68.7	55.5
SVAMP	18.0	43.3	42.3

Perturbation



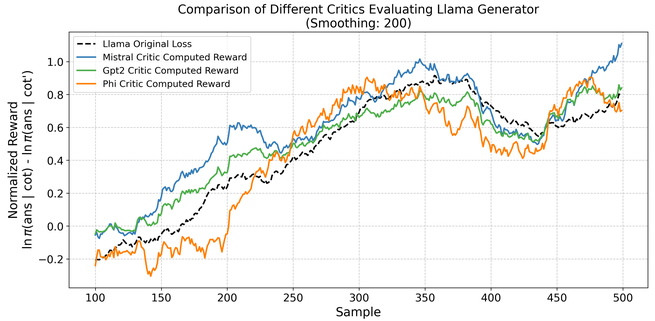
Fragility: ($\widetilde{\text{CoT}}$ = corrupted CoT)

$$\Delta = \underbrace{\ln \pi_\theta(\text{ans}|\text{CoT}) - \ln \pi_\theta(\text{ans}|\widetilde{\text{CoT}})}_{\text{Markovian}} - \underbrace{\ln \pi_{\theta'}(\text{ans}|q, \text{CoT}) - \ln \pi_{\theta'}(\text{ans}|q, \widetilde{\text{CoT}})}_{\text{Non-Markovian}}$$

Positive Δ = Mkv relies more on intact CoTs.

Cross-Model Interpretability

CoTs trained by Llama transfer to Mistral, Phi, and GPT-2 as critics. Transfer to GPT-2 (124M) argues against steganography.



As training progresses, all critics' rewards increase in tandem: CoTs encode **transferable reasoning**, not model-specific artifacts.

Qualitative Example: Wikipedia Continuation

Task: condition on 200 tokens of Wikipedia, generate 50 tokens of CoT, predict next 100 tokens.

Input passage (200 tokens): "Apoptosis (from . . .) is a form of programmed cell death that occurs in multicellular organisms and in some eukaryotic, single-celled microorganisms such as yeast. Biochemical events lead to characteristic cell changes (morphology) an"

Markovian CoT (after training): "understanding the underlying cellular processes is crucial. Compressed text: Apoptosis is a form of programmed cell death occurring in multicellular organisms and some eukaryotic microorganisms. Biochemical events lead to cell changes an Predicted next 50"
→ **Compresses key content** into the CoT bottleneck.

Baseline CoT (before training): "The text is written in a formal and technical style, which may make it difficult for some readers to understand. . . "
→ **Irrelevant metacommentary** about writing style—no useful information for continuation.

QA Perturbation Fragility

Accuracy drop (Markovian − Non-Markovian) under CoT corruption. Positive = Markovian relies more on CoT.

Dataset	CharRep	Delete	DigRep	TruncB	TruncF	Avg
ARC	+0.320	+0.424	−0.004	+0.069	+0.439	+0.250
SVAMP	+0.154	+0.204	+0.081	+0.076	+0.046	+0.112
GSM8K	+0.059	+0.069	−0.013	+0.105	+0.044	+0.053
MMLU	+0.056	+0.124	+0.004	+0.038	−0.001	+0.044
Arithmetic	−0.016	−0.003	−0.043	+0.001	−0.016	−0.015
Overall	+0.115	+0.164	+0.005	+0.058	+0.102	+0.089

Wikipedia (nats, not accuracy): avg $\Delta \ln P = +0.603$, confirming even stronger Markovian reliance.

ARC shows the clearest QA fragility (+25 pp). **Arithmetic** is a near-tie: both models treat every digit as load-bearing, so perturbation sensitivity is maximal for both.

Scope & limitations:

- Claim is **informativeness**, not full faithfulness
- Model could compute answer early and emit post-hoc CoT
- Fragility + cross-model transfer are converging evidence, not proof

Code: `github.com/scottviteri/MarkovianTraining** **Models:** Llama 3.1 8B, Mistral 7B, Qwen3 4B, Phi 3.5, GPT-2 **Tasks:** GSM8K, MMLU, SVAMP, ARC, Arithmetic, Wikipedia